Marcelino Fernando

Robotics Software Engineer · Control, Perception & Physical AI

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Summary

Robotics software engineer specializing in Physical AI — robot foundation models, vision-language-action (VLA) systems, and the compliant control that lets them act safely on real hardware. Strong focus on sim-to-real transfer, building the perception, learning, and control stacks that move robots from simulation, through hardware-in-the-loop testing, to deployment on bimanual and humanoid platforms—with depth in tactile-aware manipulation and whole-body model-predictive and impedance control.

Technical Skills

Languages	Python, C++, MATLAB
Robotics & Sim	ROS2, Isaac Sim / Isaac Lab, MuJoCo, sim-to-real & HIL testing
Control	MPC, Cartesian & whole-body control, impedance control, LQR, PID, CBF
ML / AI	PyTorch, Deep RL, VLM / VLA models, foundation models, imitation learning
Perception	OpenCV, object detection & segmentation, sensor fusion, visuo-tactile sensing
Platforms	Unitree H1 humanoid, bimanual manipulators, NVIDIA Jetson (edge deployment)
Tooling	Docker, Git / GitHub

Experience

Graduate Researcher — Sep 2024 – Present

Intelligent Space Robotics Laboratory (ISR Lab), Skoltech · Moscow

Embodied multimodal perception and robot learning for contact-rich manipulation.

- Built the Isaac Sim digital twin and synthetic data-generation pipeline behind *HapticVLA*, enabling contact-rich manipulation of fragile and deformable objects with a tactile-aware policy that requires no tactile sensing at inference time.
- Developed *GenerativeMPC*, a VLM-RAG-guided whole-body MPC framework with virtual impedance for bimanual mobile manipulation, linking semantic reasoning to compliant physical interaction; accepted at IEEE SMC 2026.

Stack: VLA models, multimodal & tactile-aware learning, RL, MPC, Isaac Sim / Isaac Lab, MuJoCo, ROS2

Engineer Intern — Sep 2025 – Mar 2026

Sber Robotics Center, R&D Department (Sberbank) · Moscow

Large-scale dataset curation for vision-language-action foundation models.

- Built large-scale manipulation datasets (10,000+ demonstrations) to train vision-language-action foundation models.
- Teleoperated bimanual manipulators and humanoid robots using master-manipulator and wearable interfaces.

Stack: Bimanual manipulation, humanoid robots, teleoperation, imitation learning, ROS2, VLA training

Research Intern — Jun 2025 – Aug 2025

MTS AI Research Division · Moscow

Real-time Cartesian model-predictive control for anthropomorphic systems.

- Designed real-time Cartesian MPC for the Unitree H1 humanoid, holding control latency under 10 ms.
- Tuned bimanual manipulation to a 95% task-success rate through trajectory planning and collision avoidance.
- Ran hardware-in-the-loop testing that cut the sim-to-real gap by 60% via systematic calibration.
- Awarded *Best Industrial Immersion Project 2025* (Skoltech).

Stack: MPC, Cartesian & real-time control, MuJoCo, hardware-in-the-loop testing, ROS2

Education

MSc with Honours in Engineering Systems — Robotics — 2024 – 2026

Skolkovo Institute of Science and Technology (Skoltech) · Moscow

- GPA 4.0/4.0. Graduated with Honours; Certificate of Achievement for *Academic Excellence*.
- Thesis: *VLM-RAG-guided Whole-Body MPC with Virtual Impedance for Bimanual Mobile Manipulator.*
- Coursework: Generative AI, Advanced Control Methods, Control Systems Engineering, Robotics, VR/AR & Haptics, Experimental Data Processing. Language of instruction: English.

BSc with Honours in Mechatronics and Robotics 2018 – 2022

National Research University "MPEI" · Moscow

- GPA 4.88/5.00. Diploma of Merit (2022); Best Conference Paper Award (2023).
- Thesis: *Mobile Mecanum Robot with Built-in Color Vision System for Object Tracking.*

Selected Projects

GenerativeMPC 2025 – 2026

Skoltech ISR Lab

- Unified three-layer framework for dual-arm mobile manipulation that bridges semantic reasoning and compliant physical interaction. VLM-RAG retrieves prior task-relevant episodes to set impedance gains, velocity limits, and safety margins, driving whole-body MPC at 10 Hz and virtual impedance–admittance control at 50 Hz.

HapticVLA 2025

Skoltech ISR Lab

- Developed the Isaac Sim digital twin and synthetic data-generation pipeline for tactile-aware VLA learning, enabling contact-rich manipulation of fragile and deformable objects without tactile sensors at inference.

Humanoid Dance Motion Imitation via RL 2025

Skoltech

- Built an Isaac Lab RL pipeline for Unitree H1 motion imitation with custom rewards and curriculum learning, reproducing a 50 s dance clip without falls at 12.1 cm root-position error.

Publications

[1] M. J. Fernando et al. (2026). *GenerativeMPC: VLM-RAG-guided Whole-Body MPC with Virtual Impedance for Bimanual Mobile Manipulation.* IEEE Int. Conf. on Systems, Man, and Cybernetics (SMC), Bellevue, WA, USA. [Accepted; Scopus & WoS] arXiv:2604.19522

[2] M. James et al. (incl. M. J. Fernando) (2026). *AnyGoal: Vision-Language Guided Multi-Agent Exploration for Training-Free Lifelong Navigation.* Conf. on Robot Learning (CoRL), Austin, TX, USA. [Submitted; No. 2 in Robotics, Scopus & WoS] arXiv:2606.13878

[3] K. Gubernatorov et al. (incl. M. J. Fernando) (2026). *HapticVLA: Contact-Rich Manipulation via Vision-Language-Action Model without Inference-Time Tactile Sensing.* IEEE/RSJ IROS, Pittsburgh, PA, USA. [Accepted; Core A] arXiv:2603.15257

[4] Y. Mahmoud et al. (incl. M. J. Fernando) (2026). *SafeHumanoid: VLM-RAG-driven Control of Impedance for Humanoid Robot.* ACM/IEEE HRI, Edinburgh, UK. [In print; Core A*] arXiv:2511.23300

[5] Fernando, M. J., Saypulaev, G. R., Saypulaev, M. R. (2025). *Analysis of a Four-Legged Robot Kinematics during Rotational Movements of Its Body.* Advanced Engineering Research (Rostov-on-Don), 25(1), 14–22. doi:10.23947/2687-1653-2025-25-1-14-22

[6] Saypulaev, G. R. et al. (incl. Fernando, M. J.) (2025). *Dynamic Analysis of a Quadruped Robot.* 7th IEEE REPEE Int. Conference, Moscow, Russia. [Oral Presentation]

[7] Fernando, M. J. et al. (2024). *Kinematic Analysis of the Translational Motions of a Quadruped Robot.* 6th IEEE REPEE Int. Conference, Moscow, Russia. [Oral Presentation]

[8] Fernando, M. J. (2023). *Development of a Mobile Mecanum Robot with Built-in Color Vision System for Tracking a Moving Object.* 29th Int. Scientific Conference, Moscow, Russia. [Best Paper Award]

Honours & Awards

Certificate of Achievement for Academic Excellence — Skoltech	Jun 2026
Academic Excellence Scholarship — Skoltech Scholarship Council	Feb 2026 & Apr 2025 – Jan 2026
Scholarship for Top Paper Development — Skoltech (safe humanoid control)	Feb 2026
Best Industrial Immersion Project 2025 — Skoltech Industrial Studies Department	Oct 2025
Best Conference Paper Award — 29th International Scientific Conference	Mar 2023
Diploma of Merit — Embassy of Angola for Outstanding Academic Performance	Jul 2022

Languages

Portuguese — Native **English** — Fluent **Russian** — Fluent **Spanish** — Beginner